

Muhammad Hassan

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EDUCATION

Virginia Tech PhD in Computer Science - CGPA 4.0/4.0 Selected Graduate Coursework: Program Analysis & Software Testing, Compiler Optimizations, Translator Design Lahore University of Management Sciences Bachelor of Science in Computer Science – Major GPA 3.70/4.00	Jan 2025 - Dec 2029 (expected) Blacksburg, Virginia Sep 2019 - Jun 2023 Lahore, Pakistan
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PUBLICATIONS

C. Saumya, M. Hassan, R. Gangaraju, K. Sundararajah, M. Kulkarni.
“Taming the Hydra: Targeted Control-Flow Transformations for Dynamic Symbolic Execution.”
In Proceedings of the ACM on Programming Languages (OOPSLA), 2026. (To appear)

Muhammad Hassan, Tahir, T., Farrukh, M., Naveed, A., Naeem, A., Shaon, F., Zaffar, F., Gehani, A., & Rahaman, S.
(2023). "Evaluating Container Debloaters." *8th IEEE Secure Development Conference (SecDev), 2023.*

EXPERIENCE

Virginia Tech - Language And Compiler Design Lab Graduate Research Assistant - Developing constant-time transformations to reduce timing side-channel leakage. - Improving symbolic execution through control flow merging for faster bug detection.	Jan 2025 - Present Blacksburg, Virginia
Aerodyne Group Software Engineer - Developed a 2D/3D annotation interface for digital-twin models. - Optimized the 3D rendering pipeline, increasing frame rate from 15 FPS to 240 FPS..	Jul 2023 - Dec 2024 Lahore, Pakistan
Lahore University of Management Science Research Assistant - Developed and evaluated BloatProfiler, a benchmarking framework for Docker image debloating. - Analzed CVE mitigation, correctness and size reduction of debloated images.	Jan 2023 - Oct 2024 Lahore, Pakistan
Jobture Software Engineering Intern Created the infrastructure to collect Job listings via APIs + Web scraping and automated with cron and AWS.	Jun 2021 - Sep 2021 Remote

PROJECTS

Control Flow Merging: Implemented control flow merging to mitigate path explosion in symbolic execution (**LLVM, KLEE**). Reduced bug finding time from 20 hours to 4 hours for mid-to-large size programs and from 1 hour to a few seconds for smaller benchmarks.

Constant Time Code: Manipulating the control flow of a program to ensure that the code always runs in constant time regardless of the input. As a result, it is able to thwart time attacks for a large number of programs.

LLM-Based Debloating: Developed a pipeline that complements debloating techniques using Azure AI GPT 3.5, RAG architecture, and LLVM coverage analysis, resulting in improved generality and security of debloated applications.

Taint Analyzer: Developed an inter-procedural taint analyzer that operates at IR level to detect vulnerabilities.

ProdeBench: Benchmarking framework for automated debloating; evaluated correctness, size reduction, gadget expressivity, and debloat time.

Additional projects: Audio Sentiment Analysis (MFCC, Chroma, Mel-spectrogram, Fourier); Twitter Real/Fake Mining; De-verse (blockchain-based censorship-resistant social media app); Netflix Recommender (NLP + unsupervised); LUMSApp (Android portal: grades, deadlines, CGPA calculator, forums).

SKILLS

Programming Languages: Python, C/C++, Assembly, JavaScript, Solidity, Haskell.

Security: Network Forensics, Cryptography, Linux, Shell Scripting.

Frameworks & Libraries: LLVM/Clang, KLEE, SMT Solvers, CUDA, TensorFlow, PyTorch, OpenCV, Scikit-learn, Angular, Node, Django, React.

Other Technologies: Git, Docker, Large Language Models, Program Analysis, Vagrant, Three.js, Konva.js, SQL, PostgreSQL, Figma, Matlab, HTML, CSS.